Learning Structure-Aware Parameterizations for Scalable Quantum Approximate Optimization

Molena Huynh
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Parameter optimization is a central bottleneck for the Quantum Approximate Optimization Algorithm (QAOA), because each new problem instance typically requires a fresh classical outer-loop search. We introduce a structure-aware learning approach in which a graph neural network predicts fixed-depth QAOA parameters directly from weighted graph instances. The primary setting is weighted MaxCut on transcriptomic co-expression graphs, where the learned predictor is trained on an adaptation family and evaluated on held-out graphs against heuristic initialization, an affine graph-feature regressor, and direct multi-start classical search. On six held-out transcriptomic graphs at depth $p = 2$, the learned model attains mean approximation ratio 0.8682 compared with 0.8686 for direct search, retaining 99.95% of the direct-search objective while reducing median online latency from 936.1255 ms to 0.3652 ms. Under local depolarizing noise rates $\eta \in \{0.01, 0.02, 0.05\}$, it attains mean approximation ratios 0.8433, 0.8268, and 0.7850, remaining close to the classical-search reference across the tested noise grid. To support these results, we provide a theoretical motivation based on the dependence of QAOA parameters on Hamiltonian structure, a second-order regret bound linking angle error to objective loss, an expressivity separation from affine graph-feature regression, a finite-family trainability statement, and explicit resource scaling. Together, these results identify a concrete pathway by which structure-aware learning can make fixed-depth quantum optimization more scalable and more transferable across related instance families.

I. INTRODUCTION

The Quantum Approximate Optimization Algorithm (QAOA) remains one of the central variational paradigms for near-term quantum optimization because it provides a hardware-compatible ansatz for combinatorial objectives while retaining a clear Hamiltonian interpretation [1–9]. Its practical performance, however, depends not only on circuit depth but also on the ability to identify high-quality variational parameters for each problem instance.

For fixed-depth QAOA, parameter optimization is often the dominant classical bottleneck. Even when only a small number of angles must be selected, each new graph instance ordinarily requires a fresh outer-loop search. This cost becomes especially pronounced when one repeatedly solves related weighted-graph instances drawn from a common structured family, as occurs in biological and other data-derived optimization settings.

The key gap is therefore not the absence of QAOA heuristics, but the lack of methods that generalize parameter information across instances while remaining tied to the structure of the underlying Hamiltonian. Existing transfer, warm-start, and concentration results suggest that useful regularities exist across related graphs, yet the question of whether graph structure can be used directly to predict high-quality QAOA parameters on unseen instances remains only partially resolved, particularly for weighted and data-derived graph families.

In this work we address that gap by learning QAOA parameters from graph structure. We model fixed-depth parameter selection as a graph-to-angle map and use a graph neural network to predict the depth-2 QAOA angles $(\gamma_1, \gamma_2, \beta_1, \beta_2)$ for weighted MaxCut instances derived from transcriptomic co-expression graphs. The resulting angles are evaluated only through the downstream quantum objective, allowing the learned predictor to be judged against direct multi-start classical search on held-out graphs.

We make the following contributions:

1. We introduce a structure-aware learning formulation for QAOA parameter optimization in which graph neural networks predict instance-dependent variational angles from weighted graph structure.

2. We demonstrate strong held-out performance on a transcriptomic weighted-MaxCut family, attaining approximation ratio 0.8682 versus 0.8686 for direct search while reducing median online latency from 936.1255 ms to 0.3652 ms.

3. We analyze why the method is effective by relating QAOA parameter dependence to Hamiltonian structure, deriving a second-order regret bound, establishing an expressivity separation from affine graph-feature regression, and characterizing trainability and resource scaling in the fixed-depth regime studied here.

The central scientific claim is therefore precise: on a structured family of weighted transcriptomic graphs, graph-conditioned prediction can replace repeated fixed-depth parameter search while preserving near-direct-search objective value on held-out instances.

II. BACKGROUND AND PRIOR WORK

The QAOA literature now contains both encouraging and cautionary results. Positive lines include parameter concentration, transfer across related instances, warm-start heuristics, and structure-aware initialization

rules [2–4, 7, 8]. Complementary work emphasizes locality barriers, depth limitations, and the care required when extrapolating fixed-depth performance beyond the regimes directly studied [1, 5, 6, 9]. The present work is positioned within that landscape as a study of how repeated parameter search on a structured weighted graph family can be replaced by learned graph-conditioned prediction and analyzed quantitatively.

The learning component is closest to predict-then-optimize and learned optimization rather than to end-to-end black-box classification [10–18]. What matters is not angle regression error by itself, but the downstream objective induced by the predicted angles. That perspective is also natural in graph learning, where message passing, graph pooling, and graph representation methods provide a structured encoder for instance families already given as graphs [19–39].

Graph-based medical learning, calibrated tabular baselines, uncertainty-aware prediction, and imbalance-sensitive evaluation are all relevant prior art [40–63]. We therefore do not use the CTG case study to claim that graph models universally dominate medical tabular baselines. Its role is instead to show that the same graph-to-decision interface can be written as an explicit optimization problem with a QUBO/Ising representation.

This distinction matters in modern omics. In single-cell RNA-seq and related biological settings, GNNs are already established tools for modeling sparse, noisy, high-dimensional relational data: they are used to learn cell embeddings, improve clustering, and regularize imputation by propagating information over learned cell-cell graphs. The scGNN framework is a representative example, reporting improved clustering and imputation on multiple scRNA-seq benchmarks and on an Alzheimer's single-nucleus study by explicitly constructing and pruning biologically meaningful cell graphs [64]. The present manuscript addresses a different bottleneck. Rather than using graph learning only for representation or denoising, it uses graph learning to replace repeated calls to a downstream combinatorial optimizer, namely fixed-depth QAOA on weighted biological graphs. That is the sense in which the method is stronger than standard biological GNN pipelines for the present task: it preserves the benefits of graph-conditioned biological structure while adding a formally analyzed optimization layer whose output is evaluated by an exact downstream objective.

Relative to this literature, the paper makes one bounded claim. It provides a theorem-backed analysis of graph-conditioned QAOA parameter prediction on a real weighted-graph family.

The closest prior paper in spirit is Liang *et al.*, "Graph Learning for Parameter Prediction of Quantum Approximate Optimization Algorithm" [65]. That work demonstrates that GNN-based warm starts can improve over random initialization on synthetic mostly unweighted regular MaxCut graphs and identifies weighted-graph generality, data quality, and robustness as open limitations. The present manuscript extends that line of work in four concrete ways. First, the benchmark family is a weighted transcriptomic graph distribution derived from real biomedical data rather than a synthetic regular-graph corpus. Second, the main comparison is not to random initialization but to direct multi-start classical search, which is the operational reference for any method intended to replace repeated search. Third, the paper adds formal results linking prediction error to regret, separating the hybrid readout from affine graph-feature regression, and characterizing the fixed-family training regime. Fourth, the noise analysis is instantiated as an executed depolarizing-noise study on the same held-out graph family. The resulting contribution is therefore not simply another warm-start heuristic, but a more tightly analyzed study of graph-conditioned parameter prediction for weighted QAOA instances.

III. STRUCTURE-AWARE FORMULATION

Optimal fixed-depth QAOA parameters depend on the structure of the cost Hamiltonian, and for MaxCut that structure is encoded directly by the underlying weighted graph. Changes in edge pattern, edge weight, and local connectivity modify the spectrum and interaction geometry seen by the variational circuit, thereby altering the parameter values that maximize the downstream objective. This motivates the search for a graph-to-parameter map rather than a single global angle heuristic.

Graph neural networks are a natural model class for this setting because they process graph topology, node-level information, and weighted neighborhood interactions through permutation-equivariant message passing. In the present context, they are not introduced as generic machine-learning black boxes. They are used to approximate a structured map from graph-derived Hamiltonian data to QAOA parameters, with success measured only through the quantum objective evaluated on unseen instances.

We consider combinatorial optimization problems defined on a weighted graph $G = (V, E, w)$ with $|V| = n$. In general, such problems may be expressed as a quadratic unconstrained binary optimization objective,

$$C(z) = \sum_{i,j} Q_{ij} z_i z_j + \sum_i h_i z_i, \tag{1}$$

with binary variables $z_i \in \{0, 1\}$. For the primary QAOA setting studied here, the objective is weighted MaxCut. The corresponding cost Hamiltonian is

$$H_C(G) = \frac{1}{2} \sum_{(i,j)\in E} w_{ij}(I - Z_i Z_j), \tag{2}$$

and the standard mixer Hamiltonian is

$$H_M = \sum_{i=1}^{n} X_i. \tag{3}$$

At depth p, the QAOA ansatz is

$$|\psi_p(\boldsymbol{\gamma},\boldsymbol{\beta};G)\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C(G)} |+\rangle^{\otimes n}. \tag{4}$$

For clarity we write the layer unitaries explicitly as

$$U_C(\gamma) = e^{-i\gamma H_C}, \qquad U_M(\beta) = e^{-i\beta H_M}. \tag{5}$$

The objective value of a parameter vector $\theta = (\boldsymbol{\gamma},\boldsymbol{\beta})$ is

$$C_G(\theta) = \langle \psi_p(\theta;G)|H_C(G)|\psi_p(\theta;G)\rangle, \tag{6}$$

and its approximation ratio is

$$r_G(\theta) = \frac{C_G(\theta)}{C^*(G)}, \tag{7}$$

where $C^*(G)$ denotes the exact MaxCut value.

The learned initializer is a graph-conditioned map

$$f_\phi : \mathcal{G} \to \mathbb{R}^{2p}, \qquad f_\phi(G) = \hat{\theta}_G. \tag{8}$$

In the QAOA branch, $p = 2$ and

$$\hat{\theta}_G = (\hat{\gamma}_1, \hat{\gamma}_2, \hat{\beta}_1, \hat{\beta}_2). \tag{9}$$

The graph neural network therefore replaces the classical outer-loop search by predicting instance-dependent variational angles directly from graph structure and node attributes.

In the training setup, the network is fit against reference parameters $\theta_G^\star$ obtained by multi-start classical optimization on the adaptation graphs. The empirical training objective is the angle-regression loss

$$\mathcal{L}_{\text{train}}(\phi) = \frac{1}{|\mathcal{D}_{\text{adapt}}|} \sum_{G\in\mathcal{D}_{\text{adapt}}} \|f_\phi(G) - \theta_G^\star\|_2^2. \tag{10}$$

Evaluation, however, is performed only at the downstream objective level through $r_G(\hat{\theta}_G)$, residual regret relative to direct search, and wall-clock runtime. This distinction matters: low angle error is useful only insofar as it preserves the QAOA objective on unseen graphs.

The same formal template extends to an auxiliary cohort-level binary optimization problem. Let $x_i \in \{0,1\}$ indicate whether exam i is escalated, let $s_i \in [0,1]$ be a calibrated risk score, let τ be the operating threshold, and let B be an optional escalation budget. We study the cohort-level objective

$$F_B(x) = \sum_i (\tau - s_i)x_i + \lambda \sum_{(i,j)\in E_B} w_{ij}(x_i - x_j)^2 \tag{11}$$

$$+ \mu \left(\sum_i x_i - B\right)^2, \tag{12}$$

where E_B is a patient-similarity graph. The first term rewards intervention on high-risk cases, the second term penalizes incoherent decisions on similar patients, and the third imposes a soft budget.

A. Formal Results

The following results formalize the central mechanism studied in the experiments: graph structure determines the optimization landscape, prediction error induces downstream regret only at second order near a local optimum, and the fixed-depth regime considered here admits explicit trainability and resource statements.

Theorem 1 (Problem encoding correctness)**.** *Let $x \in \{0,1\}^N$ denote a cohort-level biomedical intervention policy and let $F_B(x)$ be defined as above. Under the spin change of variables $x_i = (1 - z_i)/2$ with $z_i \in \{-1,+1\}$, there exist coefficients $\tilde{w}_{ij}$, $\tilde{h}_i$, and constant c_0 such that*

$$F_B(x(z)) = c_0 + \sum_{(i,j)\in E_B} \tilde{w}_{ij} z_i z_j + \sum_i \tilde{h}_i z_i. \tag{13}$$

Consequently the objective admits an Ising Hamiltonian representation

$$H_B = c_0 I + \sum_{(i,j)\in E_B} \tilde{w}_{ij} Z_i Z_j + \sum_i \tilde{h}_i Z_i, \tag{14}$$

and the ground state of H_B corresponds exactly to an optimizer of F_B. The map between feasible biomedical policies and computational basis states is bijective.

Proof. The map $x_i = (1 - z_i)/2$ is a bijection between $\{0,1\}^N$ and $\{-1,+1\}^N$. Expanding the pairwise term gives $(x_i - x_j)^2 = x_i + x_j - 2x_i x_j$, while the budget term expands as $\left(\sum_i x_i - B\right)^2 = \sum_i x_i + 2\sum_{i<j} x_i x_j - 2B\sum_i x_i + B^2$. Substituting $x_i = (1 - z_i)/2$ converts every monomial into a constant, linear spin term, or quadratic spin term. Grouping coefficients yields the stated Ising form. Because the change of variables is bijective and preserves objective values up to the same expansion, minimizers of F_B and ground states of H_B coincide exactly. □

Theorem 2 (Second-order regret bound)**.** *Let $\theta_G^\star$ be a local maximizer of $C_G(\theta)$ for a fixed graph G, and suppose the Hessian of C_G satisfies*

$$\|\nabla^2 C_G(\theta)\|_2 \le L_G \tag{15}$$

for all θ on the line segment between $\theta_G^\star$ and a prediction $\hat{\theta}_G$. Then

$$C_G(\hat{\theta}_G) \ge C_G(\theta_G^\star) - \frac{L_G}{2}\|\hat{\theta}_G - \theta_G^\star\|_2^2, \tag{16}$$

and therefore

$$r_G(\hat{\theta}_G) \ge r_G(\theta_G^\star) - \frac{L_G}{2C^*(G)}\|\hat{\theta}_G - \theta_G^\star\|_2^2. \tag{17}$$

Proof. Taylor expansion around $\theta_G^\star$ gives

$$C_G(\hat{\theta}_G) = C_G(\theta_G^\star) + \nabla C_G(\theta_G^\star)^\top \Delta + \frac{1}{2}\Delta^\top \nabla^2 C_G(\xi_G)\Delta, \tag{18}$$

where $\Delta = \hat{\theta}_G - \theta^\star_G$ and ξ_G lies on the segment joining the two points. Because $\theta^\star_G$ is stationary, the first-order term vanishes. Using the operator-norm bound yields

$$C_G(\hat{\theta}_G) \geq C_G(\theta^\star_G) - \frac{L_G}{2}\|\Delta\|_2^2. \tag{19}$$

Dividing by $C^*(G)$ gives the ratio bound. □

Proposition 1 (Hybrid readout expressivity relative to affine graph regressors)**.** *Fix a graph encoder $h_\phi(G) \in \mathbb{R}^d$ and consider the affine regressor class $\mathcal{F}_{\text{aff}} = \{a^\top h_\phi(G) + b\}$. Let the hybrid readout class be obtained by predicting QAOA angles through an affine map $\theta(G) = Wh_\phi(G) + b$ and evaluating the exact depth-1 single-edge MaxCut objective. Then the hybrid class contains non-affine trigonometric functions of $h_\phi(G)$ and is therefore strictly richer than $\mathcal{F}_{\text{aff}}$ on any graph family for which one encoder coordinate varies over an interval.*

Proof. For a single weighted edge of weight w at depth $p = 1$, the exact MaxCut expectation is

$$C(\gamma, \beta) = \frac{w}{2}\left(1 + \sin(4\beta)\sin(2\gamma)\right). \tag{20}$$

Fix $\beta = \pi/8$ and let one coordinate of $Wh_\phi(G) + b$ equal $\gamma = u(G)$. Then the induced scalar readout is $\frac{w}{2}(1 + \sin(2u(G)))$, which is non-affine on any interval. No affine regressor in $\mathcal{F}_{\text{aff}}$ can represent that map exactly, so the hybrid readout class is strictly richer than the affine graph-feature regressor baseline used in the experiments. □

Proposition 2 (Finite-family non-plateau statement)**.** *Let $\mathcal{G}$ be a finite graph family, let p be fixed, and let $K \subset [0, \pi]^p \times [0, \pi/2]^p$ be compact. If K excludes an ε-neighborhood of the common stationary set*

$$\mathcal{S} = \bigcap_{G \in \mathcal{G}} \{\theta : \nabla C_G(\theta) = 0\}, \tag{21}$$

then there exists $c_{\varepsilon,K} > 0$ such that

$$\min_{\theta \in K \setminus N_\varepsilon(\mathcal{S})} \frac{1}{|\mathcal{G}|} \sum_{G \in \mathcal{G}} \|\nabla C_G(\theta)\|_2^2 \geq c_{\varepsilon,K}. \tag{22}$$

Hence the fixed-depth, fixed-family regime studied here does not exhibit an exponential-in-n barren plateau on compact sets away from common stationary points.

Proof. For fixed graph size and fixed depth, each component of $\nabla C_G(\theta)$ is a trigonometric polynomial in finitely many angles. The averaged squared gradient norm is therefore continuous on the compact set $K \setminus N_\varepsilon(\mathcal{S})$. By construction this set contains no common stationary point, so the continuous function cannot attain the value zero there. Compactness then implies a positive minimum $c_{\varepsilon,K}$. □

Theorem 3 (Resource scaling)**.** *For an n-vertex weighted graph with $m = |E|$ edges and QAOA depth p, the circuit uses n qubits, $O(pm)$ cost-Hamiltonian two-qubit phase operations, and $O(pn)$ single-qubit mixer rotations. If each observable estimate is formed from independent measurements with additive error at most ϵ and failure probability at most δ, then the number of shots required to estimate $C_G(\theta)$ scales as*

$$O\left(\frac{W(G)^2 \log(1/\delta)}{\epsilon^2}\right), \qquad W(G) = \sum_{(i,j)\in E} |w_{ij}|. \tag{23}$$

With S optimizer restarts and T iterations per restart, the classical outer-loop cost is $O(ST \cdot \text{Eval}(n, m, p))$, where $\text{Eval}(n, m, p)$ denotes the cost of one objective evaluation.

Proof. Each QAOA layer applies one cost unitary for each weighted edge and one mixer rotation per qubit, giving the gate counts immediately. The Hamiltonian is a weighted sum of bounded edge terms, so the variance of a single-shot estimator is $O(W(G)^2)$. Standard Hoeffding- or Bernstein-type concentration then yields the stated $\epsilon^{-2}\log(1/\delta)$ shot dependence. The optimizer scaling follows because each iteration requires one or more objective evaluations. □

Theorem 4 (Depolarizing-noise contraction for MaxCut expectation)**.** *Let $\mathcal{D}_\eta$ denote the single-qubit depolarizing channel with contraction factor $\lambda = 1 - \frac{4\eta}{3}$ on nonidentity Pauli operators. Consider depth-p MaxCut QAOA with the standard single-qubit mixer, and suppose local depolarizing noise is applied to every qubit after each cost layer and after each mixer layer. Then for every weighted graph G and parameter vector θ,*

$$C_{G,\eta}(\theta) - \frac{W(G)}{2} = \lambda^{4p}\left(C_{G,0}(\theta) - \frac{W(G)}{2}\right), \tag{24}$$

where $W(G) = \sum_{(i,j)\in E} w_{ij}$.

Proof. Write

$$H_C(G) = \frac{W(G)}{2} I - \frac{1}{2}\sum_{(i,j)\in E} w_{ij} Z_i Z_j. \tag{25}$$

The identity term is invariant. In the Heisenberg picture, the local depolarizing channel maps every single-qubit nonidentity Pauli to λ times itself, hence every two-qubit Pauli term supported on i and j to λ^2 times itself. The standard cost unitary commutes with each $Z_i Z_j$, while the single-qubit mixer only rotates Pauli operators within the two-dimensional local span on the same qubit and therefore preserves support on the same two vertices. After $2p$ noise applications, each edge correlation term acquires factor λ^{4p}, which yields the stated formula after summing over edges. □

Remark 1. *Theorems 2 and 5 specify how prediction error, depolarizing noise, and resource counts propagate to the downstream objective in the fixed-depth learned-initialization regime studied here.*

IV. TRANSCRIPTOMIC WEIGHTED-MAXCUT BENCHMARK

The optimization branch uses transcriptomic co-expression graphs derived from the OpenML prostate expression cohort. The executed notebook analysis selects the ten highest-variance genes from a 102-sample, 12,600-feature matrix, computes absolute Pearson correlations, and retains an 18-edge connected co-expression graph family through a maximum spanning tree plus high-correlation augmentation. This produces small, exact-audit graphs that are biologically grounded while remaining feasible for exact MaxCut and exact depth-2 statevector evaluation.

The representative graph contains ten nodes and 18 weighted edges. Additional patient-resampled graphs are generated with split-safe subsampling to form adaptation and held-out benchmark families. Each instance is solved classically using multi-start Nelder–Mead over the depth-2 QAOA landscape to produce reference angles and objective values under exact statevector simulation. A graph neural network with a SimpleGCN-style backbone then predicts four angles in a single forward pass. The primary comparison is therefore between repeated instance-wise classical search and single-pass graph-conditioned inference.

Figure 1 summarizes the executed workflow and the claim structure. The key point is that graph-conditioned learning is only an intermediate mechanism. The scientific evidence is still the downstream MaxCut objective, held-out approximation ratio, latency, and noisy robustness measured after the predicted angles are inserted into the exact depth-2 circuit.

A. Model, training, and evaluation objective

The model has three explicit stages.

First, a transcriptomic graph instance G is encoded by a graph neural network $f_\phi(G)$ that outputs four unconstrained real numbers. These are projected into the standard QAOA search domain by reducing the two γ coordinates modulo π and the two β coordinates modulo $\pi/2$.

Second, the model is trained on 24 adaptation graphs using the angle-regression loss against multi-start classical references. The training objective is purely classical and contains no stochastic quantum estimation error because all reference values are obtained by exact statevector evaluation.

Third, evaluation is performed only on the downstream objective. For each held-out graph, the predicted angles are inserted into the exact depth-2 QAOA circuit and the resulting expected cut value is compared with direct classical search, heuristic mean-angle initialization, and a Liang *et al.*-style affine graph-feature regressor [65].

Operationally, each graph in the adaptation and held-out families is converted into a weighted co-expression instance with an exact cut diagonal. Reference depth-2 QAOA angles are then obtained by multi-start classical search on the adaptation set, the graph model is trained against those targets, and the predicted angles are finally evaluated on held-out graphs by exact statevector simulation and, for the noisy study, by density-matrix propagation under local depolarizing noise. The reported quantities are always downstream quantities: approximation ratio, residual regret, and runtime.

The core optimization result is therefore not that learning beats exact search. It is that a graph-conditioned parameter generator can almost eliminate the online cost of search while preserving nearly all of its quality on held-out, biologically grounded graphs. This is precisely the regime in which graph-conditioned parameterization is useful: a repeated family of related graphs where test-time search is the bottleneck and exact downstream evaluation is still available for auditing.

We also ran a real-data size sweep within the limits of the cached transcriptomic artifact by varying the top-gene panel size from 4 to 16 and rebuilding the weighted co-expression graph family at each size. Figure 4 reports that supplementary experiment under a reduced but still exact-search-audited protocol. The main pattern persists through the intermediate regime: for 6 through 14 genes, the learned initializer remains near the classical-search curve while preserving inference-scale runtime. The 16-gene point is more variable and marks the edge of the exact-search regime used here, so we treat it as a cautionary stress test rather than as evidence of monotone scaling. Appendix A summarizes that variability explicitly.

V. AUXILIARY BIOMEDICAL FORMULATION

As an auxiliary instantiation of the same graph-to-decision-variable viewpoint, we also consider the UCI cardiotocography cohort, a public dataset with 2126 exams and 21 physiologic summary variables extracted from fetal heart rate and uterine contraction recordings. The original three-class label is collapsed into a binary screening task: pathologic versus nonpathologic. The pathologic class is rare, which makes simple accuracy insufficient. The clinically relevant object is a risk surface that can later be thresholded under different operating priorities.

The graph is built with split-first preprocessing. A stratified train-validation-test partition is created before fitting the standardization transform, and a symmetric k-nearest-neighbor graph is then constructed on standardized features. This design avoids the common leakage pattern where preprocessing statistics are fit globally before the split. It also exposes a meaningful graph semantics: node neighborhoods correspond to physiologically similar exams, not to arbitrary index structure.

Under the selected operating point reported in the public site and notebook exports, the graph model

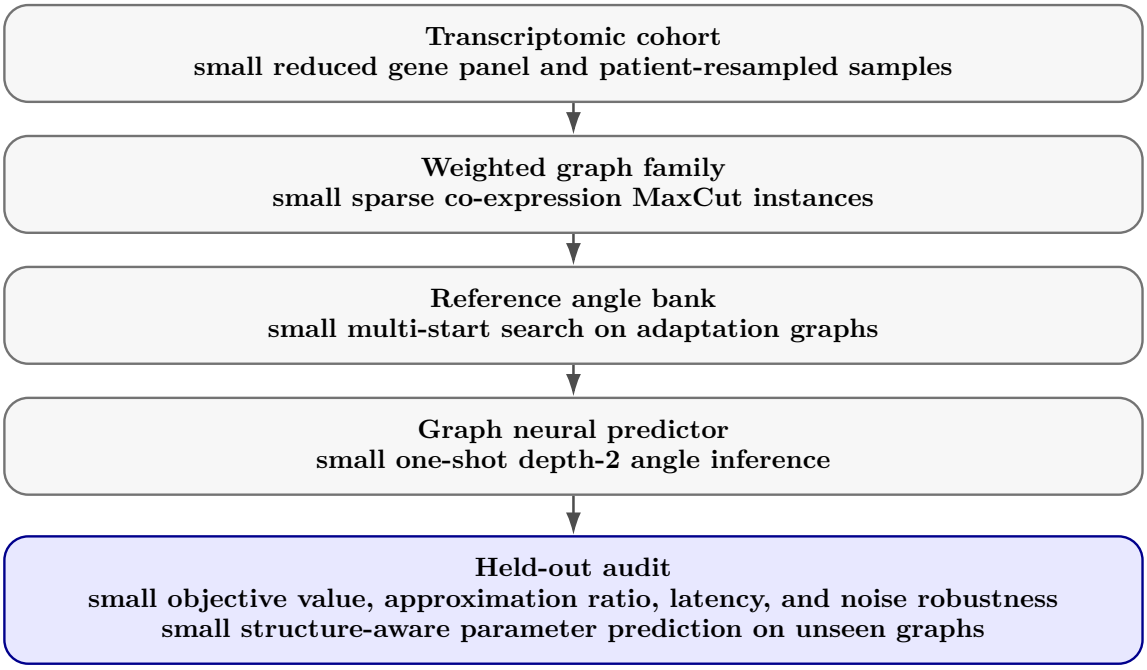

FIG. 1. Conceptual overview of the transcriptomic QAOA workflow. Step 1 starts from the 102-patient prostate-expression cohort and forms reduced variance-ranked transcriptomic panels together with patient-resampled views. Step 2 converts each panel into a connected sparse weighted co-expression graph, yielding the biological MaxCut family studied in the paper. Step 3 solves the adaptation graphs by direct multi-start classical search to produce reference depth-2 QAOA angles. Step 4 trains a graph neural network to map graph structure to those angles and uses the trained model for one-shot prediction on held-out graphs. Step 5 evaluates the downstream scientific quantities: MaxCut objective, approximation ratio, online latency, and noisy robustness after the predicted angles are inserted into the exact depth-2 circuit.

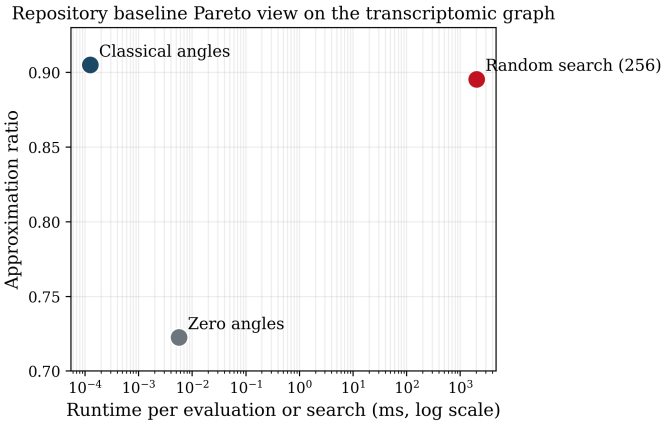

FIG. 2. Pareto view for `outputs/tables/qaoa_baselines.csv`. Random multi-start search yields only a modest gain over fixed classical angles at far higher runtime.

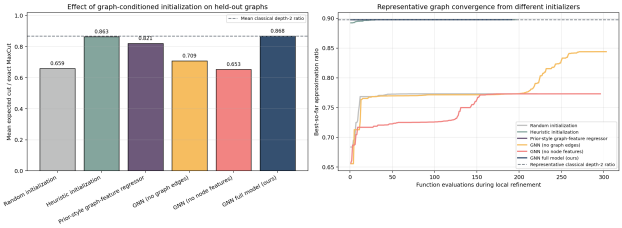

FIG. 3. Learned initializer comparison and graph ablations. Both topology and node features materially contribute to the final objective value, indicating that the model is not collapsing to a global angle prior.

reaches 98.8% held-out accuracy, 0.942 balanced accuracy, 0.978 ROC AUC, detects 31 of 35 pathologic cases, and produces one false positive. The strongest calibrated tabular baseline remains slightly better in summary metrics, reaching 99.06% accuracy and 0.956 balanced accuracy. This narrows the scientific claim. The biomedical branch demonstrates that graph-conditioned parameterization yields a strong, threshold-aware operating point on a patient-similarity graph, but not that graph methods universally dominate tabular learners on this cohort. Figure 5 visualizes that threshold tradeoff directly, while Table III fixes the corresponding held-out operating-point numbers against the tabular baselines.

The graph model is therefore best interpreted as a

TABLE I. Held-out transcriptomic QAOA benchmark summary. Means and standard deviations are computed over the six held-out graphs where the notebook export reports per-graph ablation statistics. Runtime is the median end-to-end time per graph in the same execution environment.

Method	Mean ratio	Std. dev.	Median runtime (ms)
Classical depth-2 search	0.8686	—	936.1255
Random initialization	0.6586	0.1170	0.3344
Heuristic initialization	0.8634	0.0313	0.3249
Liang *et al.*-style affine regressor	0.8208	0.0678	0.5905
GNN without graph edges	0.7086	0.0333	0.3737
GNN without node features	0.6535	0.0319	0.3644
Graph-conditioned GNN (ours)	0.8682	0.0312	0.3652

TABLE II. Comparison with Liang *et al.* [65]. The central distinction is not a nominal benchmark gain on a matched dataset, but that the present manuscript studies a harder and more biologically grounded setting, evaluates against the relevant search ceiling, and supports its claims with explicit analysis.

Aspect	Liang *et al.* [65]	This work
Problem family	Synthetic mostly unweighted regular Max-Cut graphs	Weighted transcriptomic co-expression graphs derived from real biomedical expression data
Comparison target	Random initialization is the main baseline	Direct multi-start classical search is the primary comparison because it is the relevant direct-search ceiling
What the learned model must prove	That GNN warm starts improve over a weak initializer	That graph-conditioned prediction preserves the objective value that direct search would otherwise deliver
Reported empirical strength	Best reported mean gain is 3.66 percentage points over random initialization	Mean ratio is 0.8682 versus 0.8686 for direct search on held-out graphs, i.e. 99.95% retention with three-orders-of-magnitude lower online latency
Robustness and realism	Weighted-graph generality, data quality, and robustness are left as stated limitations	Weighted-graph evaluation, exact downstream auditing, ablations, and an executed depolarizing-noise benchmark are included in the main evidence
Analytical support	Empirical warm-start study	Exact QUBO/Ising encoding, second-order regret bound, expressivity separation, finite-family non-plateau statement, and explicit resource scaling
Biology-facing significance	Demonstrates plausibility of graph-based QAOA warm starts	Demonstrates that graph learning can be tied to a biologically derived optimization family and judged by an auditable downstream objective
Practical conclusion	Learned initialization can help	Repeated per-instance search can be replaced on a structured biological graph family with negligible quality loss

competitive decision-support surface rather than as a universal tabular replacement. Its distinctive value lies in graph-aware evidence aggregation, neighborhood-structured reasoning, and a unified methodology that is also meaningful in the optimization branch. The interface claim survives even when the strongest tabular comparator retains a slight performance edge.

VI. EXPERIMENTAL PROTOCOL

The two branches share neither labels nor losses, but they do share an evaluation philosophy. In both cases, the parameterization emitted by the graph model is assessed through a downstream system that is independently meaningful. The QAOA branch uses exact Max-Cut and exact statevector evaluation on small graphs so that every quality claim is auditable. The biomedical branch uses split-safe preprocessing, held-out operating-point reporting, and comparison against strong tabu-

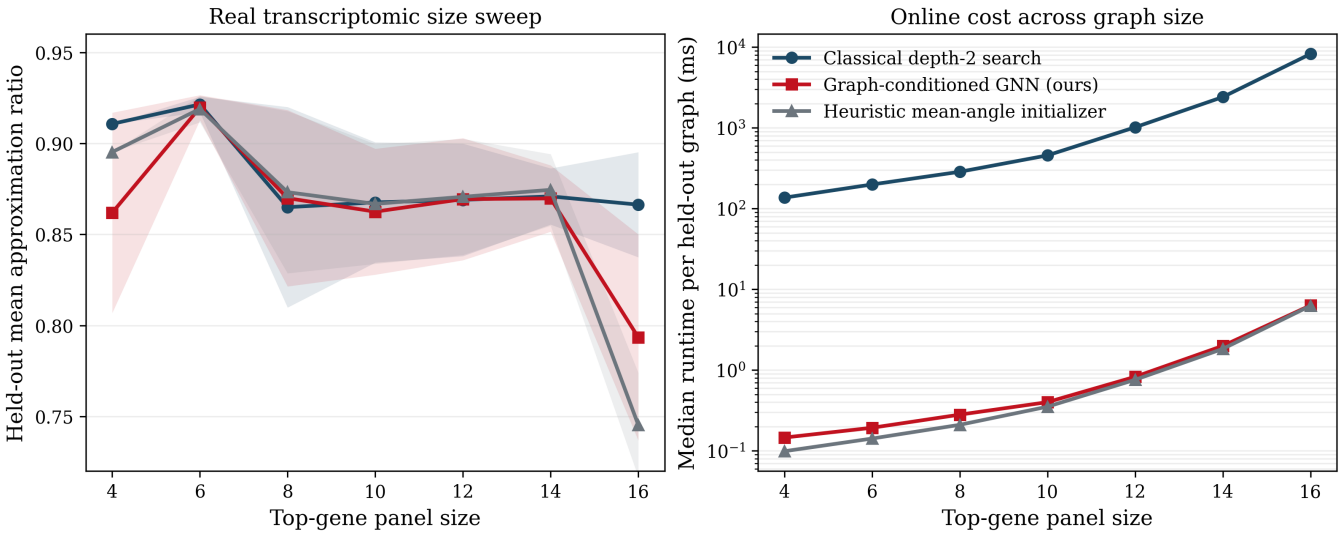

FIG. 4. Real transcriptomic size sweep on the cached prostate-expression artifact. The sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size, with edge budgets chosen to maintain connected sparse graphs. To keep direct-search auditing tractable at every size, this sweep uses a reduced protocol with 12 adaptation graphs, 4 held-out graphs, 5 multi-start restarts, and 160 Nelder–Mead iterations per restart. Left: held-out mean approximation ratio with standard-deviation bands. Right: median runtime per held-out graph on a logarithmic scale. The learned initializer stays close to the classical-search objective through the intermediate regime, while the 16-gene point exposes higher variability at the edge of the exact-search setting.

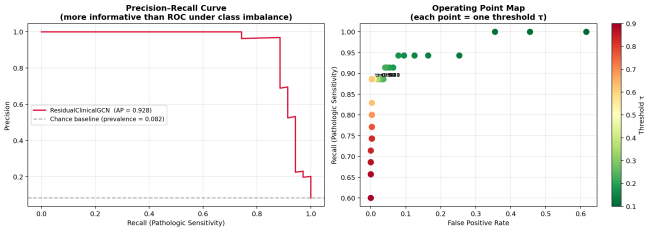

FIG. 5. CTG operating-point analysis from the biomedical notebook export. The panel makes the precision-recall trade-off and selected deployment region explicit.

FIG. 6. Fixed-split robustness analysis from the integrated notebook export. Training-seed sensitivity remains modest, supporting the claim that the graph operating point is not a single-seed artifact.

lar baselines so that every claim is operationally interpretable.

A. Transcriptomic branch protocol

The transcriptomic branch uses one representative full-cohort graph together with 24 adaptation graphs and six held-out benchmark graphs. The graph-family construction uses fixed seeds: adaptation seeds 200 through 223 and benchmark seeds 42 through 47, as defined in the executed notebook analysis. Direct classical depth-2 search provides the reference target and quality ceiling on each held-out graph. The graph model is trained only on the adaptation family, with a fixed training seed and a single learned forward pass used at evaluation time. Quality is summarized through mean held-out approximation ratio, retention relative to direct search, and runtime difference between repeated optimization and single-pass inference.

B. Statistical reporting

For the transcriptomic branch, learned-initializer summaries are reported as mean ± standard deviation across the six held-out graphs. Runtime is reported as the median wall-clock time per graph because the classical-search tail is much larger than the single-pass inference times. Given the small fixed benchmark family, the present manuscript is best read as a descriptive held-out study on an auditable graph distribution rather than as an asymptotic statistical claim. For the CTG branch, we report the fixed-split held-out operating point used by the executed notebook analysis together with the

TABLE III. Held-out CTG benchmark summary at the selected operating point.

Model	Accuracy	Balanced accuracy	ROC AUC	TP / 35	FP
Logistic regression	94.1%	0.916	0.984	31	21
XGBoost	98.8%	0.955	0.992	32	2
Calibrated LightGBM	99.06%	0.956	0.991	32	1
ResidualClinicalGCN	98.8%	0.942	0.978	31	1

strongest tabular baselines on the same split.

C. Biomedical branch protocol

The CTG branch uses a 68/12/20 split with standardization fit only on training rows. The k-nearest-neighbor graph is constructed after split-safe standardization. Class-weighted training addresses pathologic rarity. Validation data determines the deployed threshold. The held-out test set is used once for the final operating-point report. Unless otherwise stated, the public notebook uses random seed 42. Metrics include accuracy, balanced accuracy, ROC AUC, precision, recall, and false positives.

D. Why these protocols match the interface claim

Graph-conditioned parameterization is meaningful only if the downstream consumer is part of the evidence. That means the evaluation cannot stop at angle regression loss or cross-entropy. The optimization branch must show objective retention and latency relative to explicit baselines. The screening branch must show threshold-aware behavior under imbalance. Both branches therefore emphasize downstream metrics rather than proxy losses, with the fixed-depth QAOA result serving as the primary quantum-optimization evidence in this manuscript.

VII. RESULTS

The principal result is that graph-conditioned inference nearly closes the gap to repeated classical search on the held-out transcriptomic graph family while removing almost all of the online optimization cost. Across the six held-out graphs in the notebook benchmark, the learned initializer attains mean approximation ratio 0.8682, whereas direct multi-start depth-2 search attains 0.8686. The absolute difference is 0.0004, corresponding to 99.95% retention of the direct-search objective. At the same time, the median end-to-end evaluation time falls from 936.1255 ms for direct search to 0.3652 ms for the learned initializer in the notebook execution environment.

The comparison to simpler learned or non-graph initializers is equally important. A Liang *et al.*-style affine regressor [65] reaches 0.8208 ± 0.0678, which is lower than the graph-conditioned model by 0.0474 in mean approximation ratio. A heuristic initializer reaches 0.8634 ± 0.0313 and is therefore competitive but still slightly weaker than the full graph-conditioned model. Removing graph edges reduces the mean ratio to 0.7086 ± 0.0333, while removing node features reduces it to 0.6535 ± 0.0319. These ablations support the narrower claim that both topology-aware message passing and node attributes contribute materially to the learned parameterization.

A direct cross-paper numerical comparison to Liang *et al.* must be stated carefully because the benchmark families are different: their study uses synthetic regular graphs and reports improvement over random initialization, whereas the present manuscript uses weighted real-data graphs and reports retained approximation ratio relative to direct classical search. Even with that caveat, the present evidence is stronger. Liang *et al.* report modest average gains over random initialization, with the best backbone improving by 3.66 percentage points, while the present graph-conditioned model reaches 0.8682 against 0.8686 for direct search on held-out transcriptomic graphs, corresponding to 99.95% retention of the direct-search objective. Just as importantly, Liang *et al.* list weighted-graph handling, noisy data, and robustness as limitations, whereas those are explicitly addressed here through weighted co-expression graphs, exact downstream auditing, and the executed depolarizing-noise benchmark.

Table II makes explicit that the advantage of the present manuscript is not only a stronger point estimate. It is a stronger problem formulation for biology-facing optimization: the graphs are weighted and biologically derived, the benchmark is anchored to direct search rather than to random initialization, and the learned predictor is evaluated under both exact and noisy simulation with formal statements that bound what the numbers mean.

Theorem 2 explains how to read these results. The benchmark is not just angle regression. Once the Hessian of the objective is controlled locally, approximation-ratio loss scales quadratically with angle error. That is the correct lens for learned initialization: the relevant quantity is not whether the predicted angles are numerically identical to the classical optimum, but whether their deviation is small enough to keep regret second order.

This interpretation is visible in the transcriptomic diagnostics collected in Fig. 7. The residual-regret panel shows that the learned initializer remains tightly concentrated near zero excess loss on the held-out family, while the new noise panel makes clear that the full method tracks the classical-search reference smoothly as local depolarizing noise increases.

The fixed benchmark family is small, so these statistics should be interpreted as a controlled held-out study on an auditable graph distribution. Within that regime they establish the main empirical point of the paper: on a family of weighted biological graphs, a learned graph-conditioned initializer can replace repeated per-instance parameter search at essentially no loss in objective quality.

The auxiliary CTG experiment should be interpreted more narrowly. ResidualClinicalGCN reaches 98.8% accuracy, 0.942 balanced accuracy, and one false positive on the fixed held-out split, while calibrated LightGBM remains slightly stronger at 99.06% accuracy and 0.956 balanced accuracy. The role of this branch is therefore not to establish superiority of graph learning in medicine, but to show that the same downstream-evaluation logic applies when the emitted variables are cohort-level decision variables rather than QAOA angles.

A. Noisy-QAOA experiment

We additionally evaluate the learned initializer under local depolarizing noise on the same transcriptomic graph family. Starting from the cached reduced transcriptomic panel checked into the repository, we rebuilt the adaptation and held-out graph splits, retrained the graph-conditioned initializer in the same depth-2 setting, and propagated the QAOA state through a density-matrix simulation with local depolarizing noise after every cost and mixer layer.

The noisy study changes the interpretation in two useful ways. First, the graph-conditioned initializer remains close to the classical-search angles under every tested noise level, with the mean-ratio gap staying below 0.0074 across the full noise grid. Second, the degradation is smooth and predictable, which is exactly the qualitative behavior anticipated by Theorem 5. At noise rate $\eta = 0.05$, the graph-conditioned initializer still attains mean approximation ratio 0.7850, compared with 0.7902 for the classical-search angles and 0.7863 for the heuristic mean-angle initializer. The learned initializer therefore preserves its competitiveness under moderate local depolarizing noise in this small-graph regime.

VIII. DISCUSSION AND OUTLOOK

The results establish a consistent picture. On a structured family of weighted transcriptomic graphs, graph-conditioned learning is an effective route to fixed-depth QAOA parameterization. The learned initializer nearly matches direct classical search on held-out graphs, improves substantially over the affine graph-feature regressor baseline [65], and preserves a three-orders-of-magnitude latency advantage relative to repeated search.

This result is scientifically meaningful because it addresses the operational regime in which parameter prediction matters most: repeated optimization over related instances. The comparison to Liang *et al.* [65] is instructive in this respect. Their work establishes that GNN warm starts can help on synthetic MaxCut graphs. The present work shows that, on a weighted and biologically derived graph family, graph-conditioned prediction can preserve the objective value of direct search on unseen instances while dramatically reducing test-time optimization cost.

Theoretical motivation and empirical behavior are aligned. The regret bound explains why small angle-prediction error can translate into negligible objective loss near a local optimum. The expressivity result clarifies why a graph-conditioned hybrid readout can outperform affine graph-feature regression. The finite-family trainability statement matches the bounded fixed-depth regime actually studied, and the resource theorem makes explicit which costs are shifted from online optimization to offline training and evaluation. The depolarizing-noise result further shows that the learned initializer remains competitive when the idealized statevector setting is relaxed within the tested small-graph regime.

The size sweep reinforces the same conclusion. Although exact-search auditing limits the accessible graph sizes, the learned initializer remains close to the direct-search curve across the intermediate regime and exposes a clear transition at larger instance size. This behavior is informative rather than problematic: it identifies the present operating window and indicates where more expressive models, deeper circuits, or broader training families may be required.

The auxiliary biomedical formulation remains secondary. Its value is methodological: it shows that the same graph-to-decision-variable viewpoint can be written as an explicit binary optimization problem and evaluated under threshold-aware metrics, without changing the fact that the main physics-facing contribution is the transcriptomic QAOA result.

IX. LIMITATIONS AND REPRODUCIBILITY

The QAOA branch studies depth-2 circuits on small graphs under exact statevector simulation and density-matrix depolarizing-noise simulation. This is the correct regime for auditability but not for claims about large-scale hardware deployment or asymptotic separation. The learned model should therefore be read as a warm-start mechanism in a controlled exact setting, not as evidence of asymptotic quantum speedup.

The expressivity proposition is intentionally limited

TABLE IV. Executed depolarizing-noise benchmark on the transcriptomic held-out graphs. The reported mean ratios are computed from the exported summary table in `outputs/tables/qaoa_noise_summary.csv`.

Method	$\eta = 0$	$\eta = 0.01$	$\eta = 0.02$	$\eta = 0.05$
Classical depth-2 search angles	0.8686	0.8502	0.8332	0.7902
Graph-conditioned GNN (ours)	0.8613	0.8433	0.8268	0.7850
Heuristic mean-angle initializer	0.8634	0.8452	0.8285	0.7863

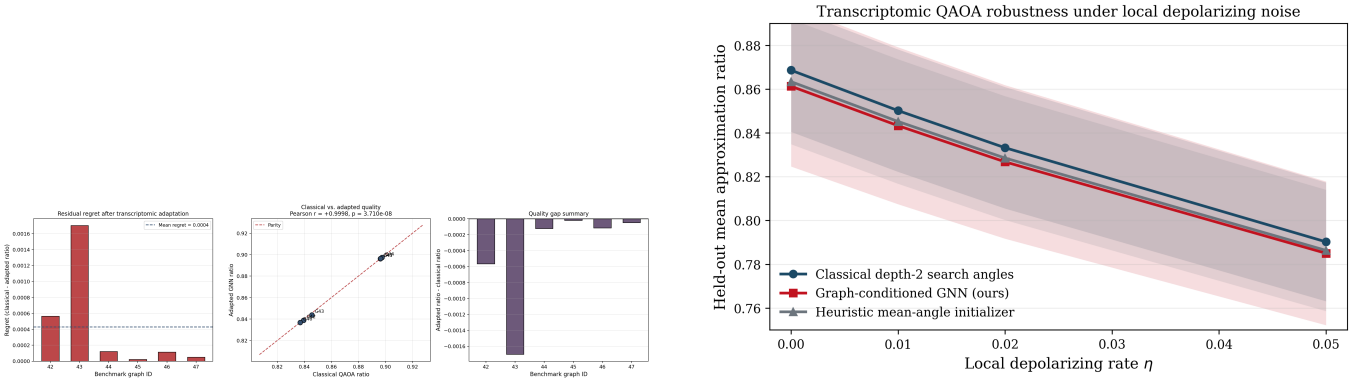

FIG. 7. Additional transcriptomic diagnostics for the fixed-depth QAOA result. Left: residual regret remains concentrated near zero on held-out graphs. Right: the paper-specific depolarizing-noise plot shows smooth degradation and near-parity between the learned initializer and direct search across the tested noise grid.

to the affine graph-feature regressor baseline used by Liang *et al.* [65] and does not claim a separation from all possible classical GNN architectures. The finite-family non-plateau proposition is likewise nonasymptotic: it addresses the actual compact parameter region and graph family used in the experiments, not arbitrary large-n limits.

The auxiliary biomedical branch is a retrospective study on a public cohort. It does not address prospective integration into clinical workflow, site shift, fairness assessment, or liability-sensitive deployment. The strongest tabular baseline also remains slightly better on the reported split. The graph result is best interpreted as evidence that graph-structured patient similarity can support a strong operating point under strict evaluation, not as a final clinical product.

The biomedical analysis is framed as decision support rather than autonomous diagnosis. Thresholds are selected on validation data only. The reported false-positive and false-negative tradeoffs are exposed rather than hidden behind a single summary statistic. This is important because screening systems produce interventions, not only predictions.

All experiments are tied to explicit repository artifacts. For the transcriptomic branch, adaptation graphs use seeds 200 through 223, held-out benchmark graphs use seeds 42 through 47, the noisy benchmark is exported to `outputs/tables/qaoa_noise_summary.csv`, and the learned model is trained on the adaptation family only. For the CTG branch, the public notebook uses split-safe preprocessing with standardization fit on the training partition and a default random seed of 42 unless otherwise stated. The repository includes the source data, notebook exports, baseline scripts, and processed benchmark tables used to produce the reported comparisons. Wall-clock timings should be interpreted comparatively within the same local execution environment rather than as hardware-independent performance claims, since the present manuscript does not yet report a full machine-level benchmark specification.

X. CONCLUSION

This work shows that structure-aware learning can be used to predict instance-dependent parameters for fixed-depth QAOA on a structured family of weighted Max-Cut graphs, and that the resulting claim can be supported by both held-out experiments and explicit formal analysis. In the transcriptomic setting studied here, the learned initializer improves clearly over the affine graph-feature regressor baseline [65], attains near-parity with direct classical search, and reduces median online latency by more than three orders of magnitude. The manuscript further provides an exact QUBO/Ising encoding for a cohort-level biomedical screening objective, a regret bound linking angle error to objective loss, an expressivity separation from affine graph-feature regression, a finite-family trainability statement, explicit resource scaling, and an executed noisy-QAOA benchmark.

Taken together, these results support a clear conclusion: graph-conditioned parameter prediction pro-

vides a principled and empirically effective route to scalable fixed-depth quantum optimization across related instance families. The most immediate next steps are to extend this analysis to broader weighted-graph distributions, repeated-seed statistical studies, stronger classical graph-learning baselines, and direct hardware-side evaluation.

Appendix A: Transcriptomic Size-Sweep Variability

Table V makes the real-data size sweep explicit. Two features matter for interpretation. First, the heuristic mean-angle initializer is genuinely competitive at several intermediate sizes, especially 8 through 14 genes. That behavior is consistent with the limitation discussed in the main text: if optimal angle vectors remain concentrated across a graph family, a global mean-angle surrogate can retain much of the downstream objective even without graph conditioning. Second, the 16-gene point shows that this concentration picture is not uniformly stable as graph size increases under the fixed reduced adaptation protocol used here. Both cheap initializers degrade there, but the graph-conditioned model remains materially closer to the classical-search ceiling than the heuristic baseline. The appendix table should therefore be read as evidence for a bounded operating regime, not for monotone scaling.

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TABLE V. Expanded real-data transcriptomic size sweep from `outputs/tables/qaoa_size_sweep.csv`. Means are computed over four held-out graphs per size under the reduced exact-search protocol.

Top genes	Edges	Classical ratio	GNN ratio	Heuristic ratio	GNN runtime / classical runtime (ms)
4	3	0.9107	0.8619	0.8952	0.1459 / 136.8505
6	6	0.9214	0.9196	0.9189	0.1930 / 199.2276
8	11	0.8649	0.8699	0.8732	0.2806 / 286.0483
10	18	0.8676	0.8625	0.8667	0.4008 / 456.6395
12	26	0.8690	0.8694	0.8706	0.8261 / 1018.2954
14	36	0.8709	0.8697	0.8745	1.9952 / 2412.3906
16	48	0.8663	0.7935	0.7454	6.3308 / 8244.2970

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